

Advanced Mechanics

Science

May, 2026

Short Title: Advanced Mechanics
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author MALHOURANI
Credits 5

Level: Introductory

Description of Module / Aims

This module covers motion under a central force, conservative forces and oscillations collisions. It introduces Lagrangian mechanics, special relativity and statistical physics.

Programmes

PHYI-0025 BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)

2 / 4 / M

Indicative Content

- Oscillating systems: forced harmonic oscillator (damped and driven)
- Motion under a central force: orbits in an inverse square force field, Euler-Lagrange mechanics
- Collisions and conservation laws: Rutherford scattering and interaction cross-section
- Rigid body motion: centre of mass; rotation about an axis; parallel and perpendicular axis theorems; the energy equation
- Kinetic theory: Maxwell and Boltzmann distributions
- Special relativity: introduction to kinematics and dynamics
- Practical programme: Experiments to measure; moments of inertia of rods and discs etc.; study of driven harmonic oscillator and resonance

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Construct and solve problems in classical mechanics.
2. Construct and solve problems in special relativity.
3. Construct and solve some problems in statistical physics.
4. Complete laboratory work and present technical reports on their work.

Learning and Teaching Methods

- Lectures.
- Tutorials.
- Laboratory work.
- Additional material in text books and other sources.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3
Continuous Assessment	30%	
<i>Practical</i>	30%	4

Assessment Criteria

- <40%:** Student has demonstrated a very limited knowledge of the course material. He/She is unable to carry out critical thinking of subject area.
- 40%-49%:** Student has demonstrated an adequate knowledge of the course material He/She is able to carry out some critical thinking of subject area.
- 50%-59%:** Shows a reasonable level of understanding of the various principles involved.
- 60%-69%:** Shows good knowledge of the area. Shows ability to reason and analyse problems in this area.
- 70%-100%:** Excellent knowledge of the subject area. Excellent reasoning skills and focused well thought out arguments.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	6	
Practical	18	
Independent Learning	87	

Supplementary Material

- Moore, D. _Introduction to Classical Mechanics, with problems and solutions_. 1st ed. USA: Cambridge University Press, 2008.
- Woolfson, M. _Resonance Applications in Physical Science_. 1st ed. London: Imperial College Press, 2015.

Requested Resources

- SCIENCE LAB: Physics